Shakunaveti Sai Sri Ram Varun

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.93
Sri Chaitanya Junior College	Class XII (TSBIE)	2023	98.40%
Sri Chaitanya School	Class X (SSC)	2021	10.00%

EXPERIENCE

ML and HPC research intern, KLA-Tencor (Jan-July 2026)

- Researched the existing literature, designed and implemented various deep learning computer vision (existing and custom) frameworks for feature extraction and improving the performance of downstream industrial anomaly detection.
- Distilled large backbone models into domain-specific models up to **100x smaller** achieving **50x faster** inference while maintaining downstream classification accuracy and analyzed various metrics validating successful distillation.
- Optimized production CUDA image processing kernels by **3-10x** through memory optimizations and kernel fusions.
- Researched on the use of **thread block clusters** and **distributed shared memory** to optimize KLA specific workloads on Blackwell B40 RTX6000 GPU.
- Worked extensively with Random Forest Classifiers, XG Boost models and various feature selection algorithms to reduce the inference time of workloads for these classifiers.

Undergraduate Research Assistant

- Currently working on machine unlearning, AI Safety for vision language models. This includes working with SOTA VLM's like qwen 3.5 9B model, and implementing fine tuning and inference passes on these large models, on multi GPU and single GPU workloads.

Undergraduate Research Assistant

- Researched and developed an automatic signal area extraction pipeline for noisy radio spectrograms in spectrum sensing, interference detection, and wireless communications.
- Implemented and analyzed hierarchical clustering algorithms, thresholding algorithms and IoU-based object matching for Signal Area Estimation.
- Benchmarked the algorithm on multiple synthetic datasets and evaluated it on metrics and opimized it iteratively.

SCHOLASTIC ACHIEVEMENTS

- Achieved AIR 883 in KVPY SA 2021
- Achieved a peak rating of 1345 in codeforces (handle: good_cow)
- Achieved a global rank of 3,325 in Meta Hacker Cup 2024

PROJECTS

IMAGE-2-LATEX

- Designed an encoder-decoder architecture to translate mathematical expression images into LATEX sequences.
- Implemented a **VGG-style CNN encoder**, 2D positional encoding, **stacked LSTM decoder**, and **Bahdanau attention** mechanism for sequence generation. Implemented **beam search decoding** during inference.

Advanced Machine Learning and Computer Vision

- Studied various research papers in the CV domain such as **De noising, Super resolution, Representation Learning, fine tuning** vision models in test-time.
- Implemented various research papers in the above domains and tried out some methods to improve the

performance. Currently started some basic research on world models such as JEPA.

Image Processing Algorithms (mini OpenCV)

- Implemented basic image processing algorithms such as blur, sharpen, convolve and edge detection from scratch in C++.
- Implemented the watershed algorithm including distance threshold and connected components analysis for image segmentation.
- Designed morphological operations such as threshold, dilation and erosion. Implemented Otsu's method for thresholding.

Machine Learning and Deep Learning Fundamentals

- Implemented a three layer **feed forward neural network** in numpy from scratch on MNIST dataset. Implemented the Forward pass, Backward pass and various optimizers, activation functions and batch wise training.
- Implemented an **LSTM** from scratch in numpy for the task next word prediction on Tiny Shakespeare Dataset. Implemented the forward pass, backward pass (**Truncated Backpropagation through Time)**, gradient clipping.
- Implemented a basic **seq2seq** model based on **bi-directional and uni-directional LSTMs** for the task machine translation in PyTorch. Achieved a 0.2 blue score for English to Spanish translation dataset of 15k sentence pairs.

SKILLS

Programming Languages: C, C++, CUDA, Python, Triton, MATLAB, Verilog

Dev tools: Docker, Git, CUDA-Toolkit(Profilers, Compute sanitizer)

Libraries: NumPy, Pandas, Matplotlib, Scipy, Pytorch, Tensorflow

RELEVANT COURSES

Math: Calculus, Differential Equations, Matrix theory, Probability and Random variables

Computer Science: Introduction to programming, Data Structures and Algorithms, Introduction to Compilers

Machine Learning and Signal Processing: Digital Signal Processing, Machine Learning, Deep Learning,

Reinforcement Learning, Speech Systems, Advanced topics in ML

EXTRACURRICULAR

- Participated in various chess and cricket events represented department and hostel in sports competitions.
- Volunteered for many events across college such as organizing marathons and teaching.
- **TA** for the course Deep Learning

POSITIONS OF RESPONSIBILITY

- **Core, Astronomy Club** Actively participated in organizing stargazing and quiz sessions.
- **Student Mentor, Sunshine Club** mentored a group of 10 students in their academics in their first year.